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Screen time, sleep duration, leisure physical activity, obesity, and cardiometabolic risk in children and adolescents: a cross-lagged 2-year study

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Abstract

Background Considering the previous research that suggested that screen time (ST), sleep duration, physical activity (PA), obesity and cardiometabolic risk factors are related, it is essential to identify how these variables are associated over time, to provide knowledge for the development of intervention strategies to promote health in pediatric populations. Also, there is a lack of studies examining these associations longitudinally. The aims of the present study were: (1) to investigate the longitudinal relationships between ST, sleep duration, leisure PA, body mass index (BMI), and cardiometabolic risk score (cMetS) in children and adolescents; and (2) to verify scores and prevalence of cMetS risk zones at baseline and follow-up.

Methods This observational longitudinal study included 331 children and adolescents (aged six to 17 years; girls = 57.7%) from schools in a southern city in Brazil. ST, sleep duration, and leisure PA were evaluated by a self-reported questionnaire. BMI was evaluated using the BMI z-scores (Z_BMI). The cMetS was determined by summing sex- and age-specific z-scores of total cholesterol/high-density lipoprotein cholesterol (HDL-C) ratio, triglycerides, glucose, and systolic blood pressure and dividing it by four. A two-wave cross-lagged model was implemented.

Results ST, sleep duration, and leisure PA were not associated with cMetS after 2-years. However, it was observed that higher ST at baseline was associated with shorter sleep duration at follow-up (B=-0.074; 95%IC=-0.130; -0.012), while higher Z_BMI from baseline associated with higher cMetS of follow-up (B = 0.154; 95%CI = 0.083;0.226). The reciprocal model of relationships indicated that the variance of ST, sleep time, leisure PA, Z_BMI, and cMetS explained approximately 9%, 14%, 10%, 67% and 22%, respectively, of the model. Individual change scores and prevalence indicated that cMetS had individual changes from 2014 to 2016.

Conclusion Sleep duration, ST and leisure PA were not associated with cMetS after 2 years. ST showed an inverse association with sleep duration, and Z_BMI was positively associated with cMetS after a 2-year follow-up. Finally, the

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prevalence of no clustering of risk factors increased after two years. These findings suggest the need to promote healthy lifestyle habits from childhood and considering individual factors that can influence cardiometabolic health in children and adolescents.

Keywords Schoolchildren, Physical activity, Sedentary behavior, Sleep duration, Obesity, Metabolic syndrome

Background

Cardiometabolic alterations are already evident in pediatric populations through the presence of metabolic syndrome risk factors, such as high blood pressure, dyslipidemia, obesity, and elevated glucose levels [1]. In 2020, the global prevalence of metabolic syndrome was 2.8% for children and 4.8% for adolescents [2]. The Brazilian adolescent's prevalence of metabolic syndrome in 2013–2014 was 2.6% [3]. In a Brazilian southern city, 11.3% had the presence of cardiometabolic risk score (cMetS) [4]. The high prevalence of cardiometabolic risk factors draws the attention of public health and highlights the importance of comprehending the factors that influences these conditions [2]. In this age group, it was suggested to use the cMetS by considering the clustered risk factors to evaluate the risk of developing cardiovascular diseases [5].

Cardiometabolic risk factors are influenced by a range of factors, including obesity, biology, sociodemographics, environment, and lifestyle habits [6]. Studies have indicated that excessive screen time (ST), inadequate sleep, and low levels of physical activity (PA), when adopted during childhood, can increase the risk of developing cardiometabolic alterations [7, 8]. For instance, a study conducted on Mexican children found a positive association between self-reported screen time and diastolic blood pressure after seven years of follow-up. Additionally, substituting 5% of sedentary time with moderate to vigorous physical activity (MVPA) was associated with a reduction in waist circumference [9].

Similarly, a prospective cohort study involving Canadian youth found that time spent in vigorous-intensity physical activity was associated with higher cardiorespiratory fitness, along with lower body mass index (BMI) and waist circumference two years later [10]. Another study involving youth over a four-year period revealed that sedentary behavior was linked to a higher cardiometabolic risk score, while a decrease in physical activity was unfavorably associated with changes in low-density lipoprotein cholesterol (LDL-C) and cardiometabolic risk score [11]. In addition, another study carried within the same population indicated that short sleep duration is associated with higher cMetS in the future [12]. These behaviors (ST, sleep and PA) also presented associations with each other, wherein increased ST negatively impacted sleep hours [13, 14] and BMI [14].

Childhood obesity is often seen as a warning sign for the development of cardiovascular diseases in adulthood, according to recent research [15–17]. Adiposity plays a central role in this process by contributing to insulin resistance, dyslipidemia, high blood pressure, and chronic inflammation, all of which can damage the cardiovascular system [16, 17]. Moreover, the harmful effects of adiposity are not limited to physical health, influencing mental well-being and cognitive function [16]. These negative consequences can manifest early in life, underscoring the importance of addressing childhood obesity as a public health priority [16]. A study that investigated the influence of adiposity parameters on cardiometabolic risk factors one year later, showed that baseline BMI was positively associated with blood pressure, triglycerides, insulin, and the cardiometabolic risk score [18].

Considering the previous research that suggested that ST, sleep duration, PA, obesity and cardiometabolic risk factors are related, it is essential to identify how these variables are associated over time, to provide knowledge for the development of intervention strategies to promote health in pediatric populations. Also, there is a noticeable gap in studies examining these associations longitudinally. The hypothesis of the present study is that ST, sleep duration, leisure PA, and BMI predict the cMetS after 2 years; and these variables may be related to each other in the follow-up. Therefore, the aims of this study were: (1) to investigate the longitudinal relationships between ST, sleep duration, leisure PA, BMI, and cMetS in children and adolescents; (2) to verify scores and prevalence of cMetS risk zones at baseline and follow-up.

Methods

Participants

This observational longitudinal study included 331 children and adolescents (aged six to 17 years; girls=191) from schools (public and private) in a southern city in Brazil. The evaluations were realized in 2014 (baseline) and 2016 (follow-up). The present study was conducted meeting Resolution 466/2012 of the National Health Council of Brazil and approved by the University of Santa Cruz do Sul ethics committee (Nº 4,278,679). An informed consent form was signed by the participant's parents or legal guardians. All methods were performed in accordance with the relevant guidelines and regulations.

To determine the power of the statistical model a post-hoc was performed using the semTools package for Structural Equation Models [19] within the software

RStudio. The following parameters were considered: error type 1=0.05; error type 2=0.10; degrees of freedom=49; null hypothesis of the Root mean square error of approximation (RMSEA)=0.08; alternative hypothesis of the RMSEA=0.05. A minimum sample of 280 participants was determined.

The study sample consisted of schoolchildren who participated in two phases of the "Schoolchildren's Health" study. These are school-based cross-sectional studies, started in 2004, and carried out periodically (the last phase carried out in 2016/2017), which evaluate various health outcomes in schoolchildren in the municipality. The evaluations were carried out from March to December, including all seasons of the year, except summer.

In each phase of the research, participating schools and students were randomly selected by conglomerates, respecting the population density of each region of the municipality (north, south, east, west, and center) of urban and rural areas. In 2004, in the first wave of research, a survey of the population density of primary and secondary school students in the municipality was carried out, which found a total of 20,380 enrollments, distributed across 50 schools. In both phases of the research included in this study, 25 schools participated in the evaluations. Classes were drawn from the selected schools to participate in the research, and all students from these classes were invited. Students aged 7 to 17 were included and those who were incapable of performing physical tests and responding to questionnaires, or who had a contraindication to collecting a biological sample (blood) were excluded.

Schoolchildren with data available in phases III (2014–2015) and IV (2016–2017) were included in the present study. The following inclusion criteria were adopted: (1) to present the difference of 2 years between the evaluations, and; (2) to have screen time, sleep duration, leisure physical activity, and obesity of complete data. Figure 1 shows the sample selection.

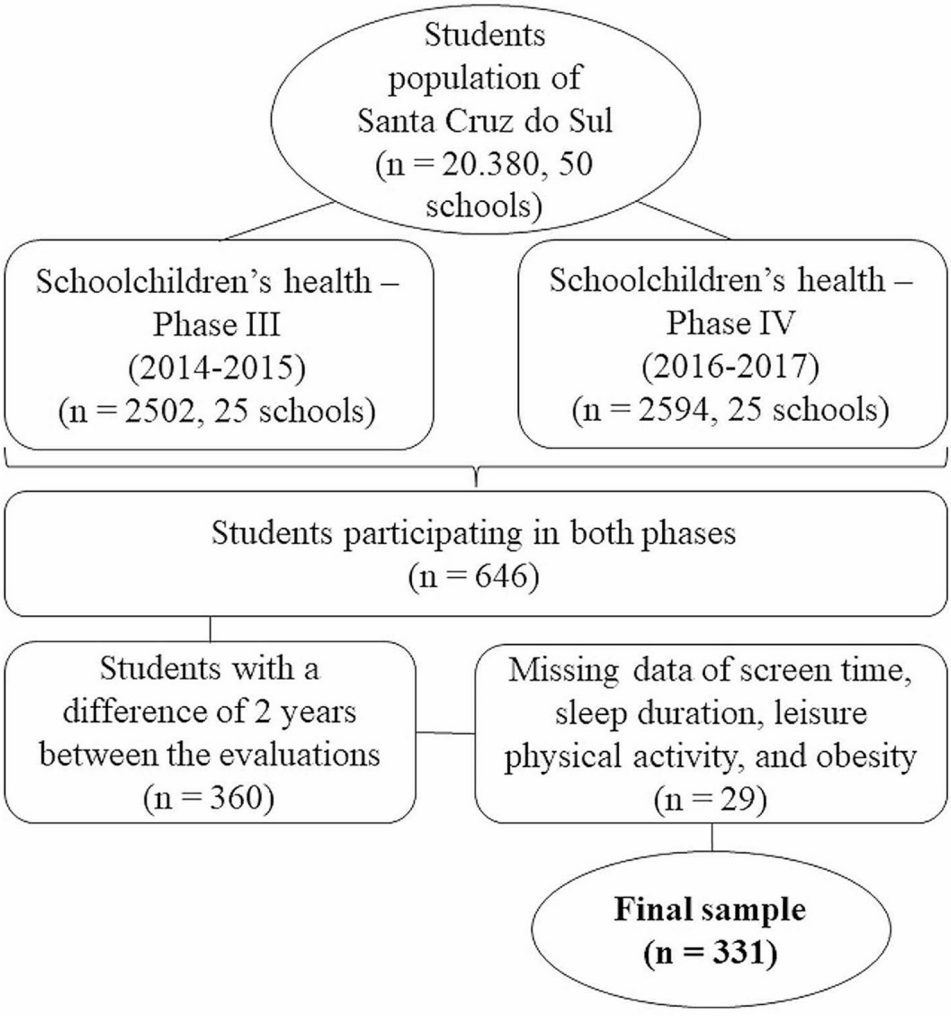


Fig. 1 Sample selection

Collection instruments

Trained professionals from the University of Santa Cruz do Sul carried out the baseline and follow-up evaluations using standardized protocols. Physical education professionals applied the questionnaire and assessed systolic blood pressure (SBP) and obesity. Blood sample collection was performed by a technical professional in nursing or a pharmacist in the Exercise Biochemistry Laboratory. To ensure consistency and accuracy of data, the same professionals conducted assessments at both time points and followed the same protocols throughout the study.

Body mass index

The present study used BMI as the obesity indicator, which was calculated using the formula: weight/height^2. These variables were evaluated by Physical education professionals using an anthropometric scale with a coupled stadiometer (Filizola®) in an anthropometric assessment room. The participants wore light clothes and were barefoot. BMI z-scores (Z_BMI) were calculated from Stavnsbo [20] considering the sex and age of the participants.

Screen time, sleep duration, and leisure physical activity

ST, sleep duration, and leisure PA were evaluated using a self-reported questionnaire adapted from the validated questionnaire by Barros and Nahas [21]. The questionnaire was answered in collection location with helped of researcher. For participants aged seven to ten years old, the questionnaire was answered by their parents.

Screen time

ST was considered to evaluated the sedentary behavior, using the following question: "How much time in minutes do you spend in front of the TV, computer, and videogame per day?". The TV, computer and videogame time were summed to obtain the total screen time (minutes/day). Tablets and smartphones were not included.

Sleep duration

Sleep duration was calculated considering the total duration (minutes per week and weekend) of sleep to obtain a mean sleep duration per day. The following questions were used: "What time do you go to sleep during the week and the weekend?" and "What time do you get up during week and weekend?".

Leisure physical activity

Leisure PA practice was determined through the sum of the total time (minutes) spent in sports or PA per week. For this, the following questions were applied: "Do you usually practice any sport/PA?" (Yes or no); "How many times a week and hours/minutes per day do you practice this sport/PA". The participant was asked to include only MVPA among the activities reported. Therefore, the physical activities included in the study are MVPA. However, as these are self-reported measures, we do not have direct intensity measures. Only leisure PA was evaluated and did not consider physical education classes.

Cardiometabolic risk factors

To determine cardiometabolic risk factors, blood samples were initially collected to measure lipid and glycemic variables, and then resting blood pressure was evaluated.

Lipid and glycemic variables

Blood samples were collected in the brachial vein from the participants after a 12-hour fast to evaluate total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), triglycerides (TG), and glucose. Collection of the blood samples was conducted by a technical professional in nursing or a pharmacist in the morning (8 a.m.) in the Exercise Biochemistry Laboratory. Serum samples were used with commercial kits (DiaSys Diagnostic Systems, Holzheim, Germany), performed on Miura 200 automated equipment (I.S.E., Rome, Italy). In addition, The TG/HDL-C ratio was calculated from the TC and HDL-C measurements using the formula: TG/HDL-C.

Systolic blood pressure

The SBP was evaluated through the auscultatory method with a sphygmomanometer and a stethoscope according to the recommended VII Guidelines of the Brazilian Society of Cardiology [22]. The assessment of SBP occurred after five minutes of sitting, using a cuff adequate according to the circumference of the arm of the participant. Two measurements were assessed, and the lowest result for SBP was used [22].

Cardiometabolic risk score

The cMetS was determined by summing z-scores of TC/HDL-C ratio, TG, glucose, and SBP and dividing it by four. The calculation of z-score considering the following equation: z-score= ([X - $\bar{x}$]/SD); where X is the continuous value observed for the risk factor; $\bar{x}$ is the predicted mean for the risk factor using regression equations from sex- and age-specific international reference values for each risk factor [20]. The international reference also provided the standard deviations (SD) used within the equations. Before analysis, TC/HDL-C ratio and TG were transformed by the natural logarithm. A healthier cardiometabolic profile is observed in more negative cMetS values. The cMetS was classified in no clustering of risk factors (values < 0.39), borderline (0.40 to 0.85), and clustering of risk factors (above 0.85) [5].

Covariables

Skin color (white, black, and brown), school type (public [state and municipality] and private), sex (male and female), and age were considered as covariables. These variables were obtained by a self-reported questionnaire elaborated by researchers.

Statistical analysis

Means and standard deviations (SD) for continuous variables and absolute (n) and relative frequency (%) for categorical variables were used to describe the sample. Data normality was tested using the Shapiro-Wilk test. The bootstrapping resampling procedure for the paired-sample two-tailed *t*-test was used to determine the statistical difference between baseline and follow-up. This procedure used a resampling of 1,000 bootstrap samples and the Bias Corrected Accelerated (BCa) method. The size of standardized differences was calculated by Cohen's d. It was considered as small difference the values of d < 0.49, a medium difference d the value between 0.50 < and < 0.79, and a large difference between the values of d > 0.80 [23].

To analyze the prevalence of high and low risk in cMetS at baseline and follow-up, individual values were classified into these risk zones using a bar-graph representation of crude cMetS for each adolescent according to the individual lower value to upper value. The prevalence was described in the same graphs at baseline and follow-up to identify any changes in data over time and observe any trends or patterns. Differences of proportions of risk were tested using the likelihood ratio with adjusted p values by the Bonferroni's method.

A cross-lagged model was implemented using Statistical Package for the Social Science (SPSS) AMOS. Cross-lagged models can analyze the longitudinal reciprocal relationships between two or more observed variables measured at two or more distinct time points. Cross-lagged models encompass both autoregressive effects, which capture the association of a variable with itself at a later point, and cross-lagged effects, which capture the association of a variable with another variable at a later point, all within the same model. In the current study, the cross-lagged model was adjusted for sex, age at baseline, school type, and skin color and there was no need to adjust Z_BMI and cMetS paths for sex and age because these values were already calculated using sex and age (see the Additional file 4 for more details about which covariate variables were included in each pathway of the model). Associations among variables at baseline were represented by correlation coefficients. Error terms were allowed to correlate within the cross-lagged model. Reporting of the results included the unstandardized regression coefficients (B) with a resampling procedure of 5,000 bootstrap samples estimating corresponding 95% confidence intervals (95% CI). The variance explained by the model for each exposure was demonstrated using the coefficient of determination (R^2). The quality of the model was assessed via the chi-square/degrees of freedom ratio ($\chi^2/df < 3.0$) [24]; via the comparative fit index (CFI > 0.90) [25]; and via the RMSEA < 0.08 [26]. It was accepted 5% type I error in all analyses.

Table 1 Descriptive statistics for ST, sleep duration, PA, obesity, and cMetS

	Mean (SD)		
	Baseline	**2-y Follow-up**	**Standardized difference**
	Mean (SD)		**Cohen's d**
Age, y	11.01 (2.24)	12.82 (2.23)	0.81*
ST, min/day	197.20 (135.48)	183.01 (151.78)	0.1
Sleep duration, min/day	572.20 (78.35)	557.48 (73.20)	0.19*
Leisure PA, min/week	107.72 (156.23)	165.85 (197.81)	0.33*
Z_BMI, kg/m^2	0.24 (0.93)	0.18 (0.96)	-
Cardiometabolic risk factors			
TC, mg/dL	159.82 (33.37)	154.43 (30.65)	0.17*
HDL-C, mg/dL	60.96 (11.51)	56.94 (10.87)	0.36*
TC/HDL-C,	2.68 (0.60)	2.79 (0.71)	-
TG, mg/dL	67.50 (31.89)	72.28 (31.09)	0.15*
Glucose, mg/dL	89.37 (9.05)	88.53 (6.91)	0.1
SBP, mmHg	104.63 (12.40)	105.62 (13.45)	0.08
cMetS, score-z	-0.10 (0.69)	-0.13 (0.65)	-
	n (%)		
Sex			
Male	140 (42.3)		
Female	191 (57.7)		
Skin color			
White	272 (82.2)		
Black	25 (7.6)		
Brown	34 (10.3)		
School type			
State	219 (66.2)		
Municipal	95 (28,7)		
Private	17 (5.1)		

SD: standard deviation; BMI: Body mass index; ST: Screen time; PA: physical activity; cMetS: Clustered cardiometabolic risk score; TC: total cholesterol; HDL-C: high-density lipoprotein cholesterol; SBP: systolic blood pressure

**Statistically difference using the bootstrapping resampling procedure for the paired two-tailed t-test ($p<0.05$)

Results

Table 1 presents the descriptive statistics of ST, sleep duration, leisure PA, obesity, and cMetS. Participants showed a small decrease in sleep duration, TC, and HDL-C mean values and a slight increase in the leisure PA and TG mean values. In addition, the sample comprises a majority of girls (57.7%), white skin color children (82.2%), and students of state schools (66.2%). The baseline analytical sample showed a higher prevalence of white skin color and a lower prevalence of municipal school type than the cross-sectional original sample. ST,

sleep duration, leisure PA, Z_BMI, and, cMetS did not differ between the baseline analytical sample and the cross-sectional sample (Additional file 1).

Figure 2 presents the individual change scores and prevalence (%) of cMetS according to baseline and follow-up data. At baseline, the prevalence of children and adolescents in the clustering of risk zones was higher than in the follow-up. Besides, the number of children and adolescents with no clustering of risk factors in follow-up was significantly higher than in baseline evaluations (Likelihood ratio [2]=6.358; $p=0.044$). Considering the graphs, it was possible to perceive that cMetS had individual crude changes from 2014 to 2016 and these results were favorable for child and adolescent health.

Table 2 presents the longitudinal associations between ST, sleep duration, leisure PA, BMI, and cMetS from baseline to follow-up in children and adolescents. Results indicated that ST ($p<0.001$), sleep duration ($p<0.001$), leisure PA ($p=0.012$), Z_BMI ($p<0.001$), and cMetS ($p<0.001$) presented significative autoregressive effects, showing stability among these variables across the 2-year time span. ST, sleep duration, and leisure PA, did not associate the cMetS after 2-years. It was observed that ST from baseline associated sleep duration at follow-up ($p=0.018$), while only Z_BMI from baseline associated cMetS from follow-up ($p=0.001$).

Data shown in Fig. 3 represents the reciprocal model of relationships between ST, sleep duration, leisure PA, Z_BMI, and cMetS from baseline to follow-up, indicating that the variance of ST, sleep time, leisure PA, Z_BMI, and cMetS explained approximately 9%, 14%, 10%, 67% and 22%, respectively, of the model. Error covariance's (correlation) coefficients at follow-up within the cross-lagged model are presented in Additional file 2. Also, standardized covariances among the variables at baseline indicated that ST and sleep duration, as well as Z_BMI, and cMetS were statistically correlated (Additional file 3). Longitudinal relationships between covariables with BMI, sleep duration, screen time, leisure PA and cMetS can be visualized in Additional file 4. Overall, the proposed model exhibited acceptable goodness-of-fit indexes: $\chi^2/df=2.313$; CFI=0.901; RMSEA=0.063 (90% CI: 0.048; 0.078; p-close=0.076).

Discussion

The findings of the present study indicated that ST, sleep duration, and leisure PA at baseline were not associated with cMetS two years later. However, there were some notable longitudinal relationships observed, including an inverse association between ST at baseline and sleep duration at follow-up, as well as a positive association between Z_BMI at baseline and cMetS at follow-up. Despite a positive association between cMetS at baseline and follow-up, the individual cMetS risk scores and prevalence differences suggest that children and adolescents' health may have improved over two years, with an increase in the prevalence of no clustering of risk factors (from 76.7 to 84.3%).

Although our results pointed to changes over time with small effect sizes in screen, sleep and leisure PA behaviors, the findings suggest an improvement in the sample's cardiometabolic profile. Considering this scenario and the great individual variability in the risk profile observed in the sample, it is expected that the associations found may differ between the variables or even not occur. The limited capacity to observe reciprocal associations in children and adolescents between the analyzed variables could be partly explained due to the variability of behaviors and health status in a real context [27, 28].

Previous longitudinal research conducted over two years has also failed to establish a relationship between lifestyle habits and cMetS [27, 28]. This suggests that two years may not be enough time to observe the influence of unhealthy lifestyles on cMetS. However, the literature indicates that adopting a healthy lifestyle at an early age is crucial to avoid the development of cardiometabolic diseases in adulthood [29, 30]. Other studies have identified associations between lifestyle variables and cMetS over time, especially with PA and sedentary behavior, indicating that higher PA and lower sedentary behavior are associated with lower cMetS over time [31, 32].

Concerning the inverse longitudinal relationship between ST and sleep duration, it was suggested that the presence and use of screen hours before sleep directly influence sleep duration, indicating negative effects on children and adolescents' health, such as somnolence, lower energy during the day, irritability, and feeling

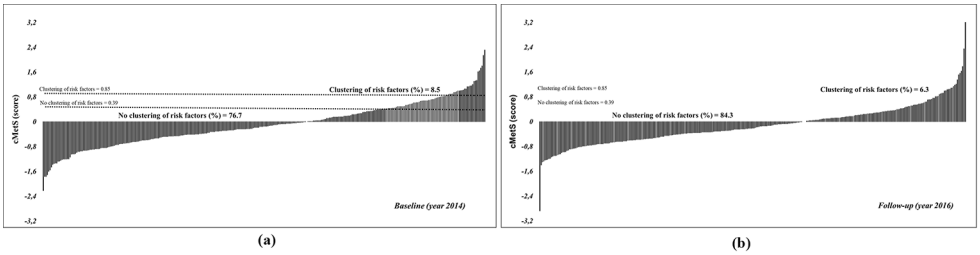


Fig. 2 Individual change scores and prevalence of cMetS according to baseline (**a**) and follow-up (**b**)

Table 2 Longitudinal associations between ST, sleep duration, PA, body mass index, and cMetS from baseline to follow-up

			B	SE	95% CI
ST (baseline)	→	ST (follow-up)	**0.269**	**0.063**	**0.148; 0.395**
Sleep duration (baseline)	→	ST (follow-up)	0.021	0.117	-0.204; 0.253
Leisure PA (baseline)	→	ST (follow-up)	0.073	0.056	-0.037; 0.180
Z_BMI (baseline)	→	ST (follow-up)	-8.820	10.038	-28.660; 10.977
cMetS (baseline)	→	ST (follow-up)	-5.889	12.280	-30.725; 17.529
ST (baseline)	→	Sleep duration (follow-up)	**-0.074**	**0.030**	**-0.130; -0.012**
Sleep duration (baseline)	→	Sleep duration (follow-up)	**0.171**	**0.048**	**0.077; 0.263**
Leisure PA (baseline)	→	Sleep duration (follow-up)	-0.024	0.028	-0.073; 0.036
Z_BMI (baseline)	→	Sleep duration (follow-up)	-4.413	4.696	-13.599; 4.896
cMetS (baseline)	→	Sleep duration (follow-up)	-0.568	6.174	-12.286; 12.007
ST (baseline)	→	Leisure PA (follow-up)	-0.008	0.076	-0.142; 0.157
Sleep duration (baseline)	→	Leisure PA (follow-up)	0.076	0.135	-0.188; 0.356
Leisure PA (baseline)	→	Leisure PA (follow-up)	**0.266**	**0.098**	**0.061; 0.450**
Z_BMI (baseline)	→	Leisure PA (follow-up)	-4.700	12.821	-29.470; 21.165
cMetS (baseline)	→	Leisure PA (follow-up)	4.472	16.633	-27.823; 37.268
ST (baseline)	→	Z_BMI (follow-up)	0.000	0.000	0.000; 0.000
Sleep duration (baseline)	→	Z_BMI (follow-up)	-0.001	0.000	-0.002; 0.000
Leisure PA (baseline)	→	Z_BMI (follow-up)	0.000	0.000	0.000; 0.000
Z_BMI (baseline)	→	Z_BMI (follow-up)	**0.835**	**0.042**	**0.754; 0.917**
cMetS (baseline)	→	Z_BMI (follow-up)	0.014	0.053	-0.090; 0.121
ST (baseline)	→	cMetS (follow-up)	0.000	0.000	-0.001; 0.000
Sleep duration (baseline)	→	cMetS (follow-up)	0.001	0.000	0.000; 0.001
Leisure PA (baseline)	→	cMetS (follow-up)	0.000	0.000	0.000; 0.000
Z_BMI (baseline)	→	cMetS (follow-up)	**0.154**	**0.037**	**0.083; 0.226**
cMetS (baseline)	→	cMetS (follow-up)	**0.290**	**0.048**	**0.199; 0.388**

Note Unstandardized regression coefficients (B), standard errors (SE), and 95% confidence intervals (CI) of each pathway included within the model; SE and 95% CI computed using percentile bootstrapping methods; Autoregressive effects of Z_BMI and cMetS reflect a standardized regression coefficient because of prior standardization; Bold denotes a statistically significant regression between variables ($p < 0.05$). BMI: Body mass index; ST: Screen time; PA: physical activity; cMetS: Clustered cardiometabolic risk score. BMI and cMetS adjusted for skin color and school type. ST, sleep duration and PA adjusted for skin color, school type, sex and age

of sadness [33]. A systematic review study found that higher ST is associated with lower sleep duration and an increase in sleep problems [13]. Furthermore, a study conducted with Brazilian adolescents found that ST is inversely related to health indicators such as BMI and sleep [14]. Thus, it is highlighted the importance of meeting recommendations for ST and sleep duration to minimize their negative effects on health indicators.

The positive relationship between Z_BMI and cMetS after two years is more commonly observed in the literature by the fact that obesity is considered the first signal of cardiometabolic disorders in children and young people and by its association with alteration in lipids, blood pressure, and glycemic indicators [15–17]. Short-term effects of adiposity on cardiometabolic health have also been reported, as decreased in body weight during childhood have been associated with better cMetS outcomes in adolescence [34]. The observed association may be related to the adoption of unhealthy lifestyle habits, including higher ST, lower PA, lower caloric expenditure, and unhealthy dietary patterns, which can lead to dysregulation of appetite hormones and accumulation of body fat [35].

Our study brings a fresh perspective to the longitudinal relationship between health behaviors and cardiometabolic risk in children and adolescents by providing individual cMetS risk scores and prevalence data. This is important because individual responses to health are complex and can vary widely, even within a population that shares similar demographic characteristics. By analyzing cMetS scores at the individual level, we gain a more nuanced understanding of how health behaviors influence cardiometabolic health outcomes in young people. Our findings suggest emphasize the importance of evaluating health behaviors in the context of both group and individual factors, taking into account the unique physiological and developmental characteristics of each child and adolescent [14, 36].

From that, it highlights the importance of meeting the recommendations of ST, sleep duration, and PA, especially in leisure time, and reducing the body weight of children and adolescents to improve cardiometabolic health in this population. Therefore, public health should promote the adoption of healthy lifestyles, emphasizing the essential role of changing lifestyle habits and involving parents and families in health promotion and prevention efforts [37, 38]. The school environment is pointed out as adequate space to carry out health actions and multicomponent interventions [38, 39].

As strengths of the study: (1) it highlights the use of cMetS, due to its utilization suggested in children and adolescent populations; (2) Longitudinal design of a representative sample of students, especially in countries with limited research funding, such as Brazil. However,

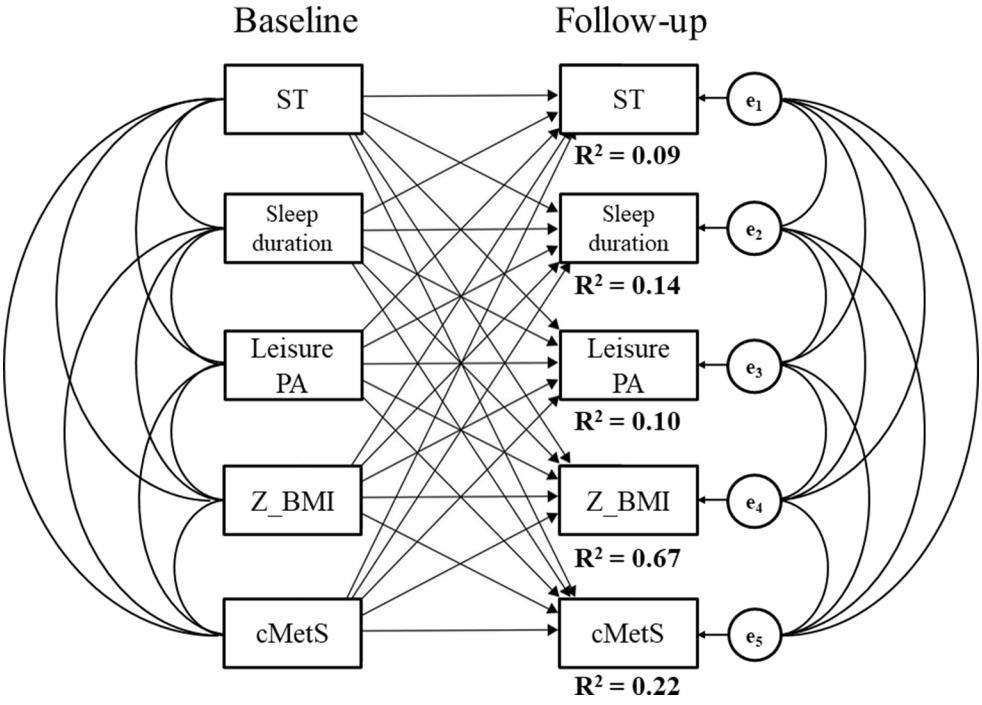


Fig. 3 Reciprocal model of relationships between the variables from baseline to follow-up. *Note* BMI: body mass index; ST: Screen time; PA: physical activity; cMetS: Clustered cardiometabolic risk score. BMI and cMetS adjusted for skin color and school type. ST, sleep duration, and PA were adjusted for skin color, school type, sex, and age

the study also acknowledges some limitations: (1) the use of adapted self-reported measurements to evaluate ST, sleep duration, and leisure PA, fact that may be subject to memory biases and social desirability; (2) it does not include other lifestyle factors, as dietary patterns; (3) PA was limited to leisure time PA and did not consider participation in physical education classes and active commuting; (4) Although the participant was asked to include only MVPA among the activities reported. The BP measurement does not allow us to assess the intensity of self-reported activities; (5) ST did not include time spent in front of tablets and smartphones; (6) The possibility of residual bias, due to unmeasured variables or measurement error cannot be ruled out.

Conclusions

Sleep duration, ST and leisure PA were not associated with cMetS after 2 years. ST was inverse associated with sleep duration after a 2-year follow-up, and Z_BMI positive associated with cMetS after a 2-year follow-up. Finally, the prevalence of no clustering of risk factors increased after two years. These findings suggest the need to promote healthy lifestyle habits from childhood and considering individual factors that can influence cardiometabolic health. In this sense, public health strategies aimed at improving cardiometabolic health in children and adolescents should involve parents and schools.

Abbreviations

95% CI	95% Confidence Intervals
B	unstandardized regression coefficients
BCa	Bias Corrected Accelerated
BMI	Body Mass Index
CFI	Comparative Fit Index
cMetS	cardioMetabolic risk Score
HDL-C	High-Density Lipoprotein Cholesterol
LDL-C	Low-Density lipoprotein cholesterol
PA	Physical Activity
R2	Coefficient of Determination
RMSEA	Root Mean Square Error of Approximation
SBP	Systolic Blood Pressure
SD	Standard Deviation
SE	Standard Error
SPSS	Statistical Package for the Social Science
ST	Screen Time
TC/HDL-C ratio	Total Cholesterol/High-Density Lipoprotein cholesterol
TC	Total Cholesterol
TG	Triglycerides
Z_BMI	BMI Z-scores

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

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Author contributions
APS, JFCS, CB, LB, and CPR participated in data organization and designed the study. APS, JFCS, CB, and VBL analyzed and interpreted the data. APS, JFCS, CB, VBL, LB, LT, KAP, LBA, RDB, and CPR contributed to the elaboration of the manuscript with critical comments about it. All authors approved the study in the current form.

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Data availability
The database used and analyzed in the present study is not publicly available as its information may compromise the participants' privacy and consent involved in the research. However, the data are available from the corresponding author, upon request.

Declarations

Ethics approval and consent to participate
The present study was conducted meeting Resolution 466/2012 of the National Health Council of Brazil and approved by the University of Santa Cruz do Sul ethics committee (Nº 4,278,679). An informed consent form was signed by the participant's parents or legal guardians. All methods were performed in accordance with the relevant guidelines and regulations.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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